An AI Copilot Helps Engineers Design Buildings

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Computers can design parts for airplanes, antennas, and race cars. But the best programs take a long time, and the human sees the finished design only at the end.

MIT researchers have built a fix: a system that works like an AI **copilot** for design. As the computer makes a **structure**, it shows the human the parts to change. **Engineers** also find it hard to trust designs that look strange.

The system uses **topology optimization**, a way of designing that finds the best shape for a **structure**. Until now, **engineers** had almost no control over how the design grew.

The researchers trained it on made-up examples of human feedback on thousands of designs.

In tests, the new **structures** were easier to make and 39 percent stronger against **buckling**, the way a **structure** bends and falls apart under pressure. It added only 15 seconds.

For now, users can only make parts thicker or thinner. The researchers hope to add more choices, and believe the system could open computer design to many more people, including students.

Word Treasure

copilot -- A helper that works alongside someone, like a second pilot.
topology optimization -- A way of finding the best shape for a structure.
buckling -- The way a structure bends and falls apart under pressure.
structure -- Something built, like a bridge or a building.
engineers -- People who design and build things like bridges and machines.

Background you need

An engineer is a person who designs and builds things like bridges, buildings, and machines.

A computer program is a set of instructions that tells a computer what to do.

Artificial intelligence is when a computer learns from examples and makes choices on its own.

Break it down

WHO: MIT researchers
WHAT: A new AI copilot for designing structures
WHERE: At MIT
WHY: Engineers could not control computer designs, and found strange-looking designs hard to trust

Why it matters

The bridges you cross and the buildings you live in are designed by engineers, and if computers help them design better, those structures can be stronger and safer for you.

Test what you remember

1. What does the new system work like?

- (A) An AI copilot for design
- (B) A 3D printer
- (C) A robot builder
- (D) A calculator

2. What design method does the system use?

- (A) Building with blocks
- (B) Drawing by hand
- (C) Topology optimization
- (D) Copying old designs

3. How much stronger against buckling were the new structures?

- (A) 15 percent
- (B) 50 percent
- (C) 100 percent
- (D) 39 percent

4. How much time did the new way add?

- (A) 15 hours
- (B) 15 days
- (C) 15 minutes
- (D) Only 15 seconds

5. What can users change for now?

- (A) Make parts thicker or thinner
- (B) Change the color
- (C) Add windows
- (D) Make parts invisible

6. Who built the system?

- (A) Doctors
- (B) High school students
- (C) MIT researchers
- (D) NASA engineers

Answer key (try the quiz first!): 1: A 2: C 3: D 4: D 5: A 6: C

Questions to think about

MY THOUGHTS (at least 20 words):